

Market Unfairness and the Compression of Bid–Ask Spreads in Cryptocurrency Option Markets

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Abstract

Put–call parity ties a European call, a put, and a matching future into a one-to-one correspondence, so on a single venue their standing quotes should be mutually consistent up to transaction costs. We propose to call the inconsistency in top of book quotes *market unfairness* (MU): the part of the quoted put–call-parity violation that remains, at the top of the book, after netting out direct transaction costs. Unlike the put–call-parity literature, which tests *trade* prices, MU is a statement about *quotes*. It is a model-free symptom of poor liquidity provision—in the spirit of the best-execution (NBBO) rules that police quote quality where identical instruments trade frictionlessly on several venues. It does not fully characterize a market’s liquidity, but it gives a clean benchmark for comparison. Using high-frequency Deribit and OKX BTCUSD and ETHUSD inverse options and matching futures over 2020–2026, we describe the level and cross-section of market unfairness and its evolution through the 2024 regulatory changes (the U.S. spot Bitcoin and Ether ETFs and MiCA). Market unfairness differs only moderately across venues and underlyings, but varies significantly over the option surface and it compressed markedly around 2024. We find evidence consistent with the entry of participants whose quoting improved market quality, as opposed to a mere increase in passive depth or tightening of spreads.

Keywords: cryptocurrency options, bid–ask spreads, market unfairness, put–call parity, limits to arbitrage, ETF approval, MiCA

JEL classification: G12, G14, G15, G23

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1 Introduction

This paper studies the top-of-book quotes of cryptocurrency option markets, and the premium that market makers require to quote them. The object is not trade prices, but standing quotes: the top of the book is a public good, and its quality is a statement about the market’s efficiency. The bid-ask spread is one of the most important frictions that impede the law of one price (Demsetz 1968, Amihud & Mendelson 1986).

There is no general theory of how wide a spread should be, but it is understood that it is compensation for the market maker to provide liquidity (Demsetz 1968, Ho & Stoll 1981, Glosten & Milgrom 1985). How much compensation is fair is a matter of what model we choose for the behavior of market makers (Ho & Stoll 1981, Bollen et al. 2004). However, there are some special cases where we *can* identify clearly unfair quotes. The best known is the case where identical instruments are quoted on several venues with no friction between transacting on one or another: this is the Nasdaq. There the best-execution rules under Regulation NMS (U.S. Securities and Exchange Commission 2005) forbid filling retail market order flow at a price worse than the National Best Bid and Offer across venues.

There are other situations which might seem similar: The ETF and ADR markets, as well as the On/Off-the-run Treasury markets (Pan & Zeng 2017, Gagnon & Karolyi 2010, Krishnamurthy 2002). But here there are real frictions (liquidity and market access) that legitimately keep the ETF market price vs ETF creation/redemption cost, or ADR and equity price, or on/off-the-run Treasury price, from being equal.

In this paper we show a situation where there are equivalent instruments traded on a single market, where price differentials should not exist at all: These are markets

in European options with matching futures. The put-call parity (PCP) relationship establishes a one-to-one correspondence between a put, a futures, and a call. The breaking of the PCP has been extensively studied in various markets (Klemkosky & Resnick 1979, Kamara & Miller 1995, Cremers & Weinbaum 2010, Ofek et al. 2004), and it has been shown that most apparent PCP breaking is due to transaction costs involved in replicating a call from a put. PCP breaking remaining beyond what can be explained by transactions costs are assumed to be idiosyncratic.

We propose to use the PCP breaking remaining in quotes after accounting for transaction costs for a new purpose: a measure of the quality of a market or venue for particular option chains. This measure we propose to call market unfairness (MU), where “unfairness” is a standardized quote-ranking label rather than a welfare judgment: it flags quotes that sit outside the cost-implied no-arbitrage band, a symptom of imperfect quote consistency. Market unfairness is not meant to characterize a venue’s liquidity completely, but it does provide a model-free measure for comparison.

Our paper will focus on characterizing U_u on cryptocurrency option markets. We describe empirical facts about market unfairness, including its response to regulatory changes in 2024. We find evidence corroborating the fact that the regulatory changes have introduced new players whose market making behavior improves quoting quality, as opposed to just introducing more liquidity.

The empirical analysis is organized around three research questions:

1. How large is U_u , and does it differ across exchanges and underlyings?
2. Did U_u drop significantly upon the 2024 regulatory and institutional regime shift?
3. Is any post-2024 drop fully explained by the usual liquidity frictions, or does it partly reflect the emergence of a new class of parity-enforcing participants?

We answer all three. Section 2 places market unfairness in the literature. Section 3 describes the market through the standard liquidity measures. Section 4 defines MMA and U_u , and describes the data and panel. Section 5 presents the level and cross-section of market unfairness. Section 6 describes the 2024 regulatory shift. Section 7 documents the compression and its robustness to the cost assumption. Section 8 reports the regressions. Section 10 concludes.

2 Related literature

2.1 *Put–call parity and crypto derivative efficiency*

Parity has long been tested, but on trade prices, not quotes.

Testing put–call parity has a long tradition in equity, index, and currency options. Early work shows apparent violations are mostly explained by bid–ask spreads, execution timing, short-sale constraints, taxes, and trading costs (Klemkosky & Resnick 1979, Nisbet 1992, Kamara & Miller 1995, El-Mekkaoui & Flood 1998, Cremers & Weinbaum 2010). Crypto options add complications: contracts are often inverse or quanto, collateral and settlement currencies differ from payoff currencies, and the reference index is not always tradable (Alexander et al. 2023, Lucic & Sepp 2024). This literature tests trade or closing prices. Our object is the standing quotes.

Recent crypto work studies efficiency; we add a quote-level measure.

Recent work treats derivative efficiency as a market-structure problem. Alexander et al.

(2024) link Bitcoin and Ether options to perpetual futures and find efficiency improving over time, especially at long maturities. Felföldi-Szűcs et al. (2024) study high-frequency parity in Binance Ether options and show violations shrink once synchronization and costs are handled. We extend this to inverse options and matching futures on Deribit and OKX, and add a comparable basis-point measure of the quoted parity gap.

2.2 *Market quality, bid–ask spreads, and the limits of arbitrage*

Arbitrage is costly, so a no-arbitrage band opens.

There is plenty of literature linking PCP breaking to bid-ask spreads. Amihud & Mendelson (1986) show spreads affect prices because investors require compensation for illiquidity. Shleifer & Vishny (1997) stress that textbook arbitrage is neither riskless nor free in practice. Gârleanu & Pedersen (2011) show how margin and funding constraints create deviations from the law of one price. All of this applies to crypto, where collateral, margin, exchange credit risk, and stablecoin conversion raise the cost of enforcing parity.

The real–synthetic quote difference splits into direct cost plus a residual.

The split of the real–synthetic quote difference into direct costs plus a residual is standard. Pontiff (1996, 2006) shows holding costs—including the idiosyncratic risk of an undiversified arbitrageur—dominate, and mispricing survives in proportion to them. Chen et al. (2021) estimate the no-arbitrage band directly from data. Gromb & Vayanos (2010) survey the theory: frictions that make convergence risky widen the band. For the market maker, Bollen et al. (2004) split the quoted spread into order processing, inventory holding, and adverse selection. Market unfairness is the analogous residual for the parity quote—the part we flag as anomalous without needing a model of fair compensation.

Such bands persist and move with market structure elsewhere.

No-arbitrage bands persist and move with market structure in other markets too. Balance-sheet constraints produced persistent covered interest parity deviations (Du et al. 2018); ETF premia persist when the underlying is illiquid or arbitrageurs are constrained (Pan & Zeng 2017, Todorov 2021).

3 **Liquidity and frictions in the crypto option market**

Standard liquidity measures gauge quote quality, one facet at a time.

Before proposing a new measure, we describe the crypto option market through the classical liquidity metrics. For each exchange and underlying we track nine channels over 2020–2026, aggregating the option surface over strike and maturity to a daily series and smoothing with a one-month rolling mean (Figure 1). Five are direct properties of the standing quotes: the average put–call bid–ask *spread* Spr (in basis points of the underlying) and the same spread in *dollars* ($Spr/10^4$ times the index level—the quoted dollar bid–ask width); the quote *depth* in dollars (the size resting at the top of book) and in underlying units (“shares”, the dollar depth divided back by the index); and the *staleness* $Stale$ (the fraction of quotes that fail to refresh within the sampling interval). We report the basis-point and the dollar (respectively share) versions side by side because a rising underlying mechanically shrinks a bp spread and inflates a dollar depth even when the quoted width or the resting share size is unchanged. The other four are dynamic illiquidity and resiliency measures, computed for each strike–maturity cell from its intraday five-minute series and averaged over the surface:

- the *Amihud* illiquidity ratio (Amihud 2002), the average of $|\text{return}|/\text{volume}$ over the cell’s five-minute call and put series—how far the price moves per unit traded—with volume measured in traded contracts (*shares*) and in *dollars* (contracts times the index);
- the *spread recovery time*, a resiliency measure—the third dimension of liquidity alongside tightness and depth (Kyle 1985, Foucault et al. 2013). After a *large* widening—the spread’s deviation from its centered rolling one-hour baseline crossing above two robust standard deviations (from the day’s median absolute deviation)—we fit an AR(1) decay to the deviation over the following hour and report its half-life $-\ln 2 / \ln \rho$ in minutes, the time for a shock to die out by half; we compute it for both the bp and the dollar spread. Conditioning on genuine widenings, in the spirit of the order-book resiliency literature (Degryse et al. 2005, Large 2007), discards the small idiosyncratic quote wiggles that otherwise leave an unconditional AR(1) near unit-root and uninformative.

The trade-based measures use the summed call and put volume; the quote-based measures need no trades.

The direct quote frictions improve sharply around 2024.

Figure 1 shows the bp spread and staleness falling markedly and dollar depth rising over the sample, with the improvement concentrated around the 2024 regulatory and institutional changes (Section 6); the Amihud ratios and the spread-recovery times are plotted on the same axes for comparison. The dollar spread and the share depth, by contrast, move much less—for Bitcoin the dollar spread even drifts up—so a good part of the bp tightening and the depth increase reflects the higher underlying rather than a genuinely narrower quoted width or more resting size. By the price-neutral yardsticks the improvement is real but more muted.

But none of these says whether the quotes are mutually consistent.

Each channel captures a single facet of liquidity, is tied to a venue’s microstructure, and—for the trade-based measures—depends on observed order flow. Comparing venues or option chains on any one of them mixes several things at once, and a market can look deep and tight while still quoting a put, a call, and a future that violate their own no-arbitrage relationship. The rest of the paper develops a complementary, model-free measure built only from standing quotes—market unfairness—that asks exactly this question of internal consistency, and then shows that it, too, fell in 2024.

4 The absolute market unfairness: concept and measurement

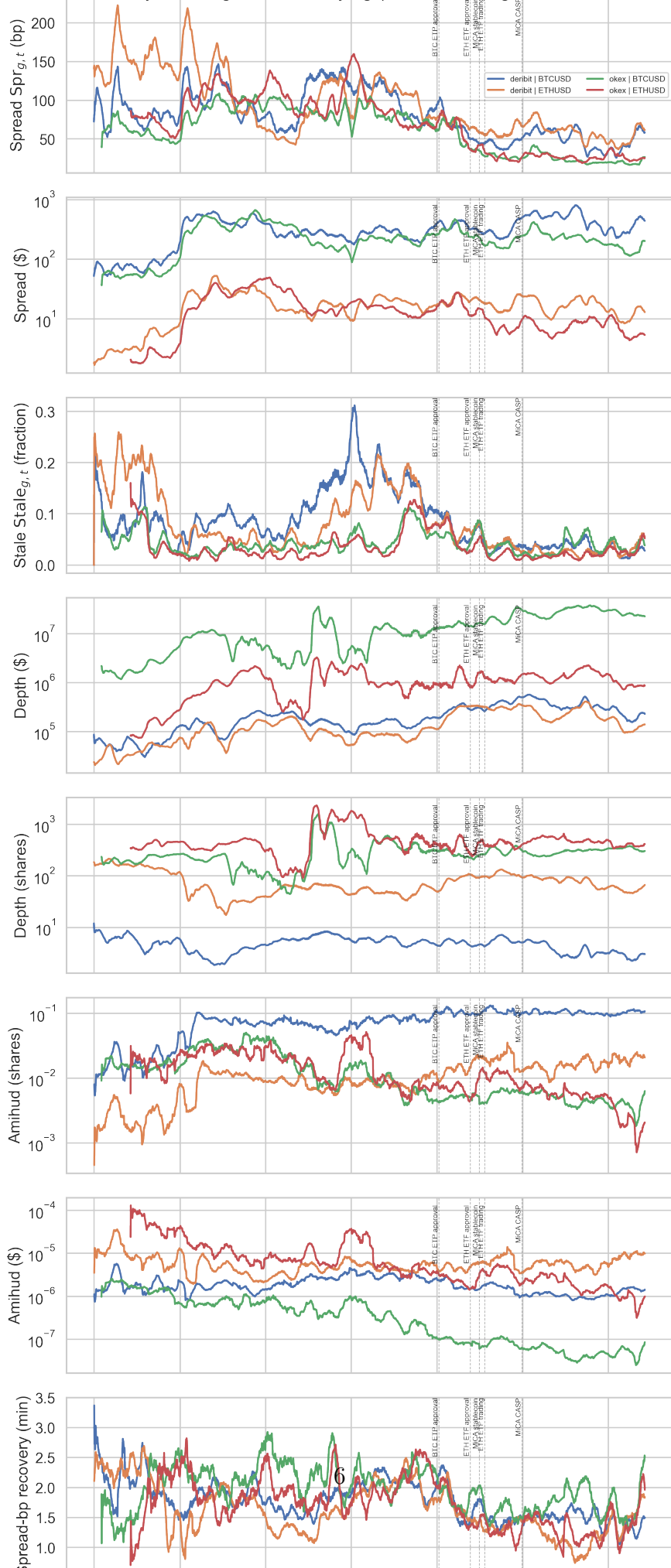
The organizing idea is the accounting identity of costly arbitrage:

$$\text{real-synthetic quote difference} \leftarrow \text{synthetic quote transaction costs} + \text{absolute market unfairness.} \quad (1)$$

U_u is the parity gap left in quotes after direct costs.

A parity deviation in the quote invites the market maker to join the unfair quote, but only if getting filled can produce riskless profit. This potential riskless profit we call market unfairness. We do not decompose the direct cost. Primarily it reflects direct trading costs imposed by the exchange, but also a risk premium (counterparty, settlement, convergence, inventory, funding, adverse selection), latency, stale books, tick-size effects, or a low chance

Friction channels over time by exchange and underlying (1-month rolling mean, over strike and maturity)



of a fill; the limits-to-arbitrage literature catalogues such frictions (Pontiff 2006, Chen et al. 2021, Gromb & Vayanos 2010), and the spread decomposition names their market-making counterparts (Bollen et al. 2004). Our claim is narrower and model-free: whatever its source, a positive U_u is a cost-adjusted parity gap left in the quotes, hence a symptom of imperfect quoting. Everything downstream is net of one direct-cost parameter (0.003 of notional); Section 7 shows the conclusions empirically do not strongly depend on the exact choice of the direct-cost parameter.

4.1 Market maker advantage and the absolute market unfairness

Parity in crypto options must be written after currency conversion.

An additional source of complexity in executing our study is the fact that the most liquid contracts are inverse. We use BTCUSD and ETHUSD options on Deribit and OKX, which are quoted against a USD index, but collateral, premium, and settlement are in coin. Parity must therefore be written after converting premia and the futures leg to a common currency (Alexander et al. 2023, Lucic & Sepp 2024): the coin-denominated premia are scaled by the conversion price S .

Four paths implement the arbitrage.

For a matched strike–expiry pair the profit depends on how the arbitrage is run: backward or forward, joining the call or the put side. By convention *forward* means arbitrage involving going *long* futures and vice versa. In every path the maker *joins* a standing quote rather than crossing it: he posts and profits only if a taker hits him. For instance, in backward/join-call-bid he buys the call at C_{bid} ; in forward/join-put-bid he sells it at C_{bid} . This gives four paths (Table 1), each net of a round-trip cost and converted to basis points by 10,000/index_level.

A tickpath is one (pair, path) observation.

Let $m_{i,p,t}$ be the signed basis-point payoff of path p for pair i at time t , after conversion and cost adjustment. We call each (pair, path) observation a *tickpath*: every synchronized quote (“tick”) for a matched pair gives four tickpaths, one per path, so a cell (g, t) with $n_{g,t}$ pairs has $4n_{g,t}$ tickpaths. Both objects below are averages of m over tickpaths; they differ only in which tickpaths enter.

Market maker advantage (MMA) is the mean of $m_{i,p,t}$ taken over *all* tickpaths in a cell (g, t) :

$$\text{MMA}_{g,t} = \frac{1}{4n_{g,t}} \sum_{i \in g,t} \sum_{p=1}^4 m_{i,p,t}, \quad (2)$$

where $n_{g,t}$ is the number of matched pairs in the cell. The average includes the many paths with no positive profit after costs, so *mean MMA is negative*: it sits near minus the round-trip cost. MMA is the outcome of a maker who joins every quote indiscriminately.

Market unfairness (MU) comes in two complementary forms, both built from the positive part of m . The *conditional unfairness* U_c averages m over only the *positive-MMA tickpaths*—those with $m_{i,p,t} > 0$:

$$U_{c,g,t} = \frac{\sum_{i \in g,t} \sum_{p=1}^4 \max(0, m_{i,p,t})}{\sum_{i \in g,t} \sum_{p=1}^4 \mathbf{1}\{m_{i,p,t} > 0\}}, \quad (3)$$

the average *size* of the wedge, given that one exists. Let the *unfairness rate* $R_{g,t}$ be the

share of the $4n_{g,t}$ tickpaths with a positive payoff,

$$R_{g,t} = \frac{1}{4n_{g,t}} \sum_{i \in g,t} \sum_{p=1}^4 \mathbf{1}\{m_{i,p,t} > 0\}, \quad (4)$$

and let the *unconditional unfairness* U_u average the same positive part over *all* tickpaths:

$$U_{u,g,t} = \frac{1}{4n_{g,t}} \sum_{i \in g,t} \sum_{p=1}^4 \max(0, m_{i,p,t}) = R_{g,t} U_{c,g,t}. \quad (5)$$

The two forms factor into rate times size.

The identity $U_u = R U_c$ is the paper’s organizing decomposition: the per-quote wedge is how *often* a wedge appears (R) times how *large* it is when it does (U_c). We read the cross-section through this size/rate split (Figures 7–8); the time series and regressions use the unconditional unfairness U_u . Both are in basis points of notional.

U_u is the expected profit of an imaginary maker who joins only the favorable side.

Read market unfairness through an imaginary maker who joins a quote only when the parity payoff is in his favor. It is an expectation, not a realized profit: whether he is filled depends on queue position and on a taker crossing. The bid–ask spread already pays him for the usual costs of making a market—order processing, inventory, adverse selection, queue risk. Market unfairness is separate: it measures only how differently the two equivalent instruments—the real quote and its parity-implied synthetic—are priced against one another. MMA, conditional unfairness, and unconditional unfairness are three summaries of the same tickpath distribution of m : its mean over all tickpaths, its mean over the positive ones, and the mean of its positive part.

Table 1: Arbitrage-profit formulas by path and contract style. To obtain the basis-point statistics used for MMA/MU, multiply each expression by 10,000/index_level. Each expression includes the trading-cost term $\text{cost} = \text{cost_per_notional} (0.003) \times \text{index_level}$ from Table 3. Contract size cancels in the basis-point expression. For inverse contracts, S_{bid} and S_{ask} are the prices at which the coin payoff is converted to USD or stablecoin.

Path	Linear contracts	Inverse contracts
fwd, join put bid	$C_{bid} - P_{bid} - (F_{ask} - K)e^{-rT} - \text{cost}$	$(C_{bid} - P_{bid})S_{bid} - (F_{ask} - K)e^{-rT} - \text{cost}$
bck, join call bid	$P_{bid} - C_{bid} + (F_{bid} - K)e^{-rT} - \text{cost}$	$(P_{bid} - C_{bid})S_{bid} + (F_{bid} - K)e^{-rT} - \text{cost}$
bck, join put ask	$P_{ask} - C_{ask} + (F_{bid} - K)e^{-rT} - \text{cost}$	$(P_{ask} - C_{ask})S_{ask} + (F_{bid} - K)e^{-rT} - \text{cost}$
fwd, join call ask	$C_{ask} - P_{ask} - (F_{ask} - K)e^{-rT} - \text{cost}$	$(C_{ask} - P_{ask})S_{ask} - (F_{ask} - K)e^{-rT} - \text{cost}$

4.2 The range of arbitrage configurations

Geometrically, U_u compares a real quote to its synthetic twin.

It is useful to see what market unfairness is measuring geometrically. For a given option leg—say the call—the quotes define a *real* market $[C_{bid}, C_{ask}]$, while parity combined

with the put and the matching futures defines a *synthetic* market $[C_{bid}^{syn}, C_{ask}^{syn}]$, where $C^{syn} = P + (F - K)e^{-rT}$. Both intervals live on the same price axis, and their relative position determines whether, and how strongly, an arbitrage exists. Figure 2 shows the three qualitatively distinct configurations.

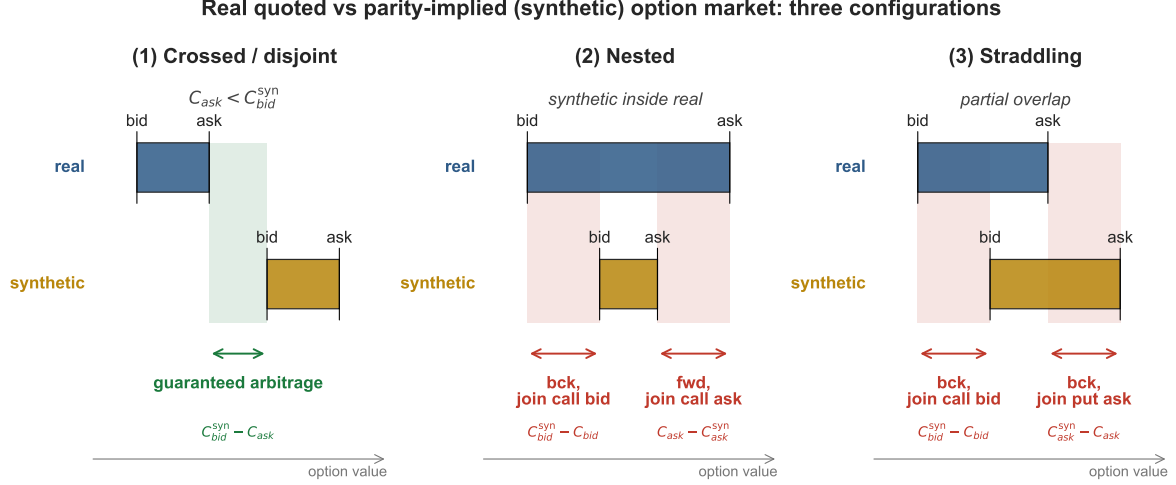


Figure 2: Real quoted versus parity-implied (synthetic) option market. When the markets are crossed and disjoint ($C_{ask} < C_{bid}^{syn}$, or the symmetric case), there is a single guaranteed riskless arbitrage. When one market is nested inside the other, or when the two straddle each other, there is no guaranteed profit but two paths carry a *potential* market-maker advantage, realized only if the maker's quote is joined and then hit. In the nested case these are the backward join-call-bid and forward join-call-ask paths ($C_{bid}^{syn} - C_{bid}$ and $C_{ask} - C_{ask}^{syn}$), with the join-put payoffs equal to their negatives (exactly so when the futures market has zero spread; the trading cost need not be zero); in the straddling case they are the backward join-call-bid and backward join-put-ask paths ($C_{bid}^{syn} - C_{bid}$ and $C_{ask}^{syn} - C_{ask}$). The latter two configurations are what market unfairness aggregates. Schematic, illustrative levels.

Only the crossed case is a guaranteed arbitrage.

In the crossed case the real ask lies below the synthetic bid (or vice versa); buying one and selling the other locks in the gap. In the nested and straddling cases there is no guaranteed profit: the payoff depends on which side is joined and whether a taker crosses. These conditional states are what MMA and U_u aggregate over the surface.

U_u is not the spread; it is where the quote sits.

MMA and market unfairness do not measure the quoted half-spread itself—the compensation makers demand for liquidity provision. They measure where the quotes *sit*: each real quote must track its synthetic counterpart within the cost cushion (Section 4.3), and market unfairness is what stands outside that tolerance. This is the through-price gap of the NBBO analogy from the introduction, with the parity-implied synthetic market as the second venue.

4.3 The no- U_u band and the shape of the MMA distribution

How far can a quote move before it is unfair?

The configurations in Figure 2 raise a sharper question: for a *given* synthetic market, how far can a real quote move before it becomes unfair? The answer pins each real quote to its

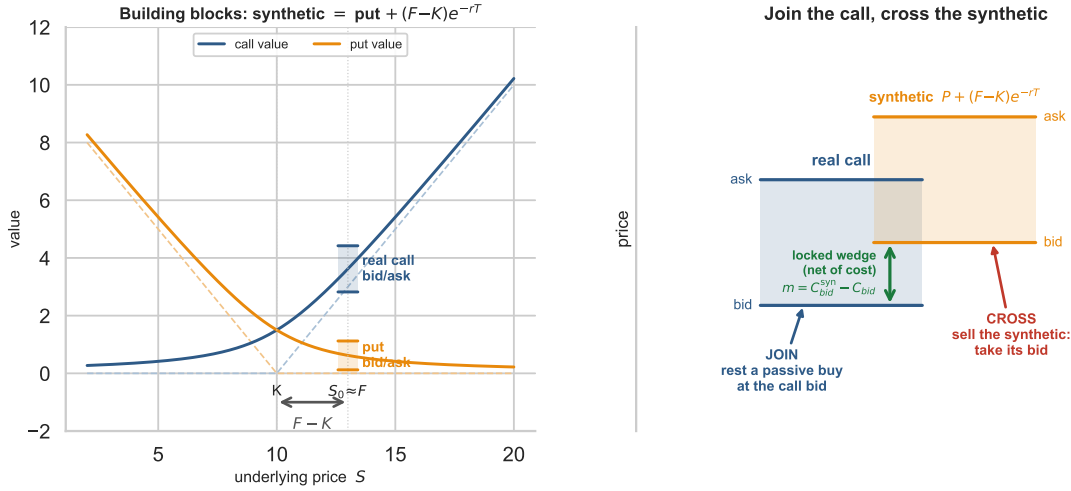


Figure 3: One path in detail: the backward join-call-bid (illustrative, exaggerated). *Left:* the synthetic call is assembled from parity—the call and put quotes and the forward-minus-strike $F - K$, giving $C^{\text{syn}} = P + (F - K)e^{-rT}$; the object we measure is a top-of-book *quote* (a bid/ask price), not a payoff. *Right:* when the real call quote sits below its parity-implied synthetic quote $C^{\text{syn}} = P + (F - K)e^{-rT}$, the maker *joins*—rests a passive buy at the real call bid—and, once that resting order is hit, *crosses* the synthetic market, selling the parity-equivalent synthetic at its bid. The locked wedge is $C_{\text{bid}}^{\text{syn}} - C_{\text{bid}}$, net of the round-trip cost (the nested backward join-call-bid of Figure 2). Only the resting leg is passive; the wedge is therefore a quoted payoff *conditional on the maker being filled*, not a guaranteed trading profit. Schematic, illustrative levels.

synthetic counterpart within a tolerance set by the trading cost, and it fixes the shape that the MMA distribution “should” take.

Two paths bound the fair call ask.

Consider the real call ask C_{ask} . Two of the four paths in Table 1 contain it—backward/join-put-ask and forward/join-call-ask. Requiring both payoffs to be non-positive (no U_u) yields

$$P_{\text{ask}} + (F_{\text{ask}} - K)e^{-rT} - \text{cost} \leq C_{\text{ask}} \leq P_{\text{ask}} + (F_{\text{bid}} - K)e^{-rT} + \text{cost}. \quad (6)$$

A fair quote must track its synthetic within a cost cushion.

Both bounds use the *same-side* put quote P_{ask} , so the band is centred on the synthetic ask at the futures mid, $C_{\text{ask}}^{\text{syn}} \equiv P_{\text{ask}} + (F_{\text{mid}} - K)e^{-rT}$, with half-width $\text{cost}' \equiv \text{cost} - \frac{1}{2}(F_{\text{ask}} - F_{\text{bid}})e^{-rT}$. The symmetric calculation pins C_{bid} to $C_{\text{bid}}^{\text{syn}}$. So a fair real market must track the synthetic market quote by quote, to within $\pm \text{cost}'$ (Figure 4). The minus sign in cost' is correct: when the futures market is very wide, cost' turns negative, meaning no fair synthetic market exists and market unfairness is undefined. This did not occur in our data.

The band’s width comes from the cost.

The role of the cost is easy to mis-state. It is *not* that C_{ask} merely lies inside the synthetic bid–ask; that would carry no cost at all. Rather, C_{ask} is tied to the synthetic *ask* within the $\pm \text{cost}'$ cushion. With $\text{cost} = 0$ and a positive futures spread, cost' is negative, so the interval in (6) inverts and is empty: no strictly fair ask exists. As all frictions vanish, the band collapses and the real market should become strictly identical to the synthetic one—the law of one price.

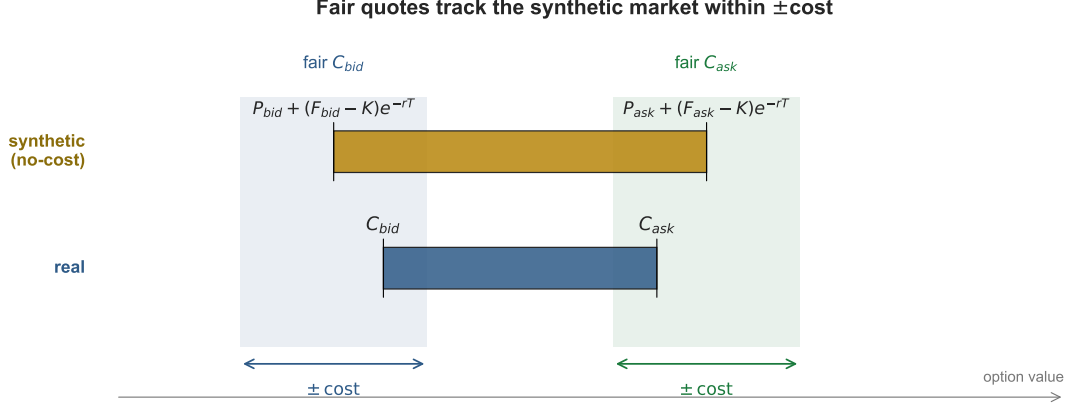


Figure 4: No U_u requires each real quote to track its matching synthetic quote within $\pm \text{cost}'$ (the half-width $\text{cost}' \equiv \text{cost} - \frac{1}{2}(F_{ask} - F_{bid})e^{-rT}$). The band centres are the frictionless (no-cost) synthetic prices $P_{bid} + (F_{bid} - K)e^{-rT}$ and $P_{ask} + (F_{ask} - K)e^{-rT}$; the cushion around each is $\pm \text{cost}'$. As frictions shrink, the real market is squeezed onto the synthetic market. Schematic, illustrative levels.

Uniform quoting would imply a flat MMA rectangle.

Equation (6) also fixes the shape of the MMA distribution. Write $u = C_{ask} - C_{ask}^{\text{syn}}$; the two conjugate C_{ask} payoffs are exactly $u - \text{cost}'$ (backward, join put ask) and $-u - \text{cost}'$ (forward, join call ask). As C_{ask} ranges over its fair band $u \in (-\text{cost}', \text{cost}')$, each sweeps linearly over $(-2\text{cost}', 0)$. If makers quoted with no preferred point inside the band, the implied MMA would be *uniform* on $(-2\text{cost}', 0)$: a rectangle of width $2\text{cost}'$ just left of zero. That rectangle is the no-information benchmark (dashed in Figure 5).

Competition instead concentrates MMA near minus the cost.

The observed distribution instead peaks sharply near $-\text{cost}'$. Two things point there, and it is worth separating them. Pooling the conjugate paths—which share the call-ask and put-ask quotes and whose payoffs sum to $-2\text{cost}'$ —makes the distribution symmetric about $-\text{cost}'$ by construction. That it is also sharply *peaked* there, rather than spread across the fair band, is the economic content: a maker who improves one side’s MMA lowers the conjugate by the same amount, so the symmetric quote concedes no advantage. Competitive quoting thus keeps the modal MMA near $-\text{cost}'$, and since the futures spread is small in our data $\text{cost}' \approx \text{cost}$ (Figures 5 and 6). Stale or aggressive quotes populate the tails; the right tail crossing zero is the U_u .

4.4 Data and sample

The sample uses high-frequency Deribit and OKX BTCUSD and ETHUSD inverse options and matching-expiry futures from Tardis.dev (Tardis.dev 2026), spanning 2020 through June 2026, with synchronized best bids and asks for calls, puts, and futures. Ticks are processed with our open-source codebases—`balazs14/TardisProject` and `balazs14/publications` (subfolder `AMU-liquidity-premium`)—which implement synchronization, filtering, parity-path construction, and market unfairness aggregation (Králik 2026b,a). The same calculation applies across the full horizon, so pre/post-2024 comparisons are coherent.

Table 2 summarizes coverage. Deribit lists a wider surface—more contracts, expirations, and strikes per day—while spreads are broadly comparable, slightly wider for ETH on

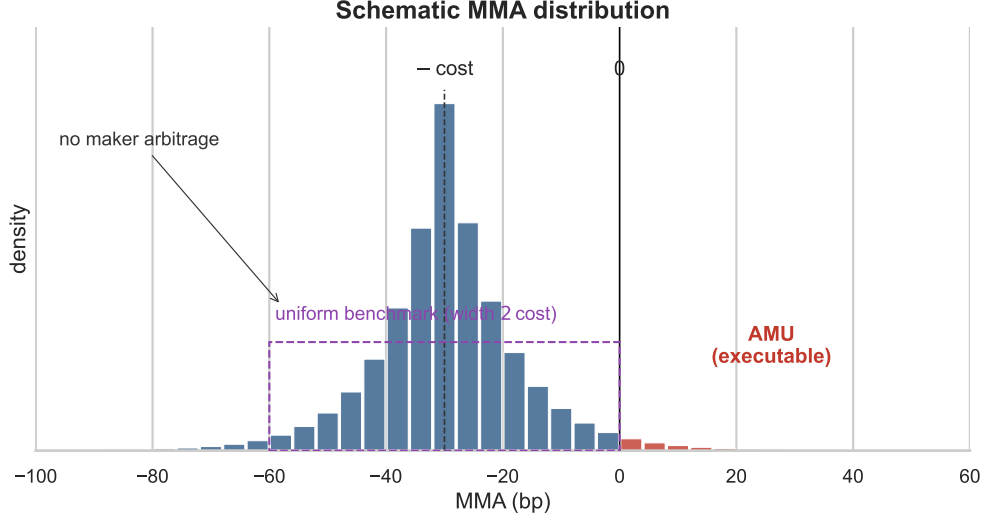


Figure 5: Schematic MMA distribution. The dashed rectangle over $(-2\text{cost}', 0)$ is the no-information benchmark implied by the no-MU band (uniform quoting), with $\text{cost}' \equiv \text{cost} - \frac{1}{2}(F_{ask} - F_{bid})e^{-rT}$. Competitive quoting instead concentrates the mode at $-\text{cost}'$, the symmetric equilibrium of the conjugate backward/forward paths; the mass that crosses zero into positive territory is the U_u . Schematic, illustrative levels.

OKX. The OKX series are longer because Deribit introduced matching-expiry futures later.

	BTCUSD		ETHUSD	
	deribit	okex	deribit	okex
Avg num contracts	70	50	42	34
Avg num expirations	6	5	6	4
Avg num strikes	42	36	29	31
Avg call spread (bp)	76	55	82	65
Avg put spread (bp)	63	55	75	62
Num days in sample	2336	2269	2336	2145

Table 2: Summary statistics of matched put–call pairs in the 2020–2026 sample.

4.5 Panel construction

Observations are synchronized on a regular time grid, filtered to matched pairs with usable quotes, and aggregated into cells indexed by (day, exchange, ref_sym, rel_strike_bucket, tte_bucket). This preserves the cross-section while supporting the time-series analysis. Table 3 collects the processing assumptions.

Table 3: Data-processing assumptions for MMA/ U_u construction.

Component	Assumption
U_u clipping rule	MMA series are clipped to $[0, 100]$ basis points before positive-part averaging.
MMA clipping rule	MMA series are clipped to $[-500, +500]$ basis points.
Spread filter	Call and put spreads are capped at 1000 basis points.
Relative-strike filter	Allowed range is $[0.79, 1.21]$.
Minimum quote size	<code>min_quote_size_dollar</code> must be at least 5000.
Subsampling	Last-known observation carried forward to the configured interval (default <code>5min</code>); a leg with a missing or non-updated bid/ask price or size is marked stale.
Stale-quote filter	disabled: when enabled, ticks with a stale call, put, or spot leg are dropped before any U_u /MMA aggregation.
Discount rate	Risk-free rate assumed $r = 5\%$; it enters only through the factor e^{-rT} . Not a crucial assumption—see the sensitivity analysis (Section 9).
Trading cost	Fixed round-trip cost of 30 bp of notional (<code>cost_per_notional</code> = 0.003). Not a crucial assumption—see the sensitivity analysis (Section 9).

5 The level and cross-section of the absolute market unfairness

5.1 *The shape of the MMA distribution and why it peaks below zero*

The MMA density peaks at minus the cost.

The most informative object in the data is the shape of the MMA distribution (Figure 6). Pooled across markets and paths, the density peaks sharply at minus the trading cost—near -30 basis points—and falls away on both sides. Zero splits the region where the maker cannot profit from the positive tail that feeds U_u .

The peak is where competition drove the wedge down to cost.

The reading is central to the paper. Makers improve quotes only as much as competition forces, so the mass sits at $-\text{cost}$: in the modal state the wedge is competed down to about the cost of the trade. Quoting is noisy, so the distribution is not a point mass. Quotes too good for the taker side fill the left tail; quotes too bad leave a positive wedge and fill the right tail. The right tail’s conditional mean is the conditional unfairness, its share is R , and their product is the unconditional unfairness. Market unfairness is nonzero because in an imperfect market market makers do not always face enough competition. While some of this is idiosyncratic noise, we will see evidence that not all of it is.

5.2 U_u across exchanges, moneyness, and maturity

Venues differ moderately; the thinnest book has the most U_u .

The global averages (Table 4) show venues differ moderately, not on a simple BTC/ETH split. The standout is ETH on OKX, the thinnest book (Table 2), as expected if market unfairness tracks quoting quality: the least deep book leaves the largest wedge above cost. Positive MMA is not rare: a large fraction of ticks carry at least one positive-MMA path,

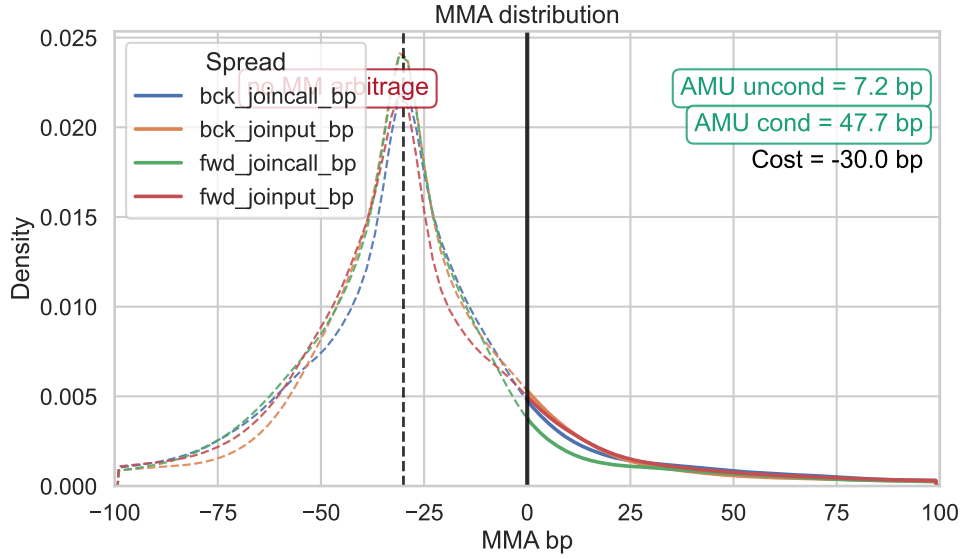


Figure 6: Average density of the four MMA paths across all markets. The distribution peaks at minus the round-trip cost (dashed line); everything to the left of zero yields no maker profit, and only the positive tail to the right of zero contributes to U_u .

so the positive-part average is a routine phenomenon, not isolated glitches. The ratio of the unconditional to the conditional column recovers R (Eq. 5).

		cond	uncond	positive MMA	num pairs
underlying	exchange				
BTCUSD	deribit	49.33	7.48	18,099,637	46,878,362
	okex	46.22	6.48	11,929,538	32,953,499
ETHUSD	deribit	46.40	7.31	11,202,380	27,995,392
	okex	47.89	7.69	8,631,085	21,137,090

Table 4: Global U_u summary across exchanges and underlyings. Market unfairness is the positive-part mean of MMA, in basis points of notional: *cond* averages over the positive-MMA paths only, *uncond* over all paths (equivalently, *cond* times the fraction of paths with positive MMA). A *path* is one of the four put–call-parity join paths (backward/forward $\times$ join-call/join-put) evaluated at a single synchronized quote; each matched put–call pair contributes four paths. The last two columns count ticks: *positive MMA* is the number of synchronized ticks with at least one positive-MMA path, and *num pairs* the number of matched put–call pair ticks.

Over the surface, U_u varies mainly through frequency, not size.

The surface shape is sharper than the venue averages and regular across all four markets. In Figure 7, the unfairness rate R (four paths per matched quote) (lower panel) traces a deep V—near one in the wings, a few percent at the money—while the conditional size (upper panel) is flat. Their product, the unconditional unfairness, inherits the V. Figure 8 is the same in maturity with a U-shaped frequency: high at the shortest maturities, minimal near one month, rising again for long-dated contracts. U_u is smallest for month-scale,

near-the-money contracts and largest in the illiquid corners. Over the surface it is the *incidence* of mis-quoting that varies, not its size.

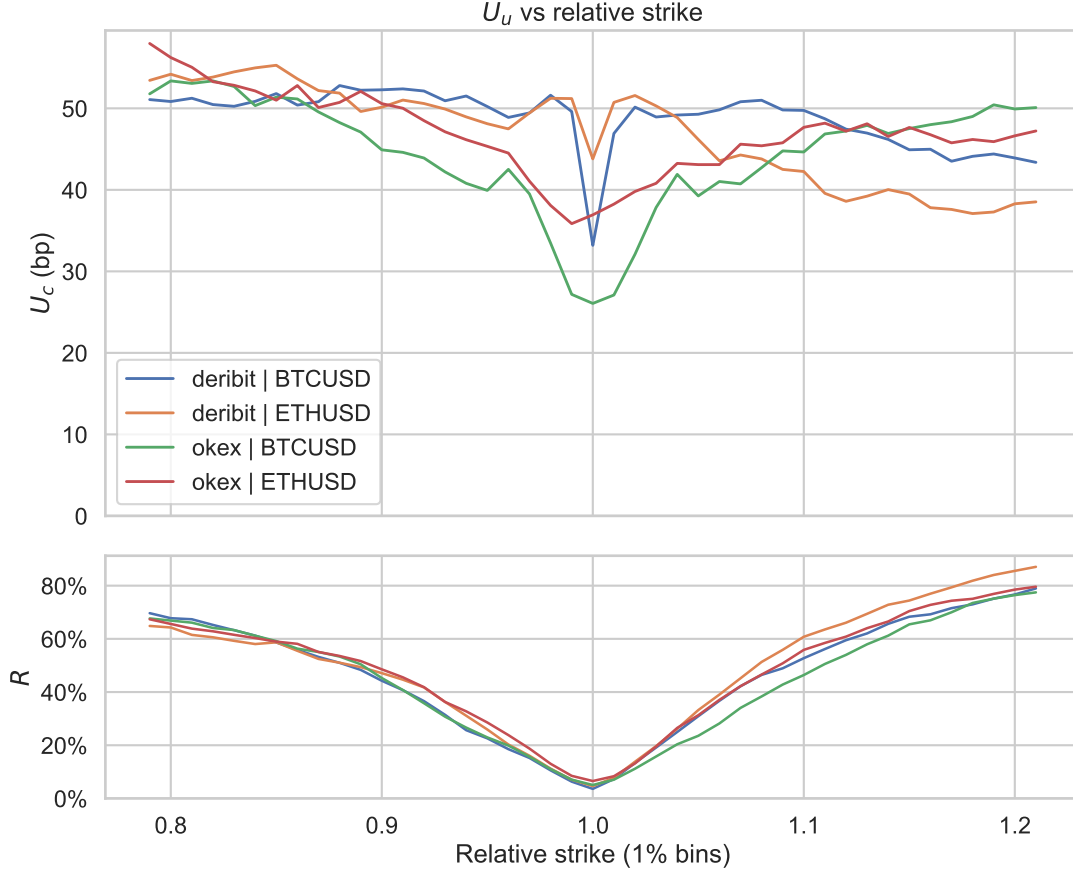


Figure 7: U_u by relative strike across exchanges and underlyings, decomposed into its two margins: the conditional size (upper panel, comparatively flat) and the unfairness rate R (four paths per matched quote) (lower panel, a deep V minimized at the money). Their product, the unconditional unfairness, is minimized at the money and rises sharply into the wings.

Position on the surface matters more than venue.

These regularities speak to the first question, on the size and cross-section of market unfairness. U_u differs across venues, but position on the surface—distance from the money and from a one-month horizon—explains far more of its variation than venue does. Both patterns line up with liquidity: the wedge is largest where books are thin and quotes stale.

6 The 2024 regulatory shift

Regulation, not parity, changed in 2024.

Three milestones cluster in 2024. On January 10 the SEC approved spot Bitcoin ETPs, connecting crypto price risk to regulated custody, authorized-participant arbitrage, and mainstream brokerage (U.S. Securities and Exchange Commission 2024b). In May it approved spot Ether ETFs, which began trading in July (U.S. Securities and Exchange Commission 2024a). In parallel, MiCA came into force in stages: stablecoin provisions in mid-2024, the CASP licensing regime at end-2024 (European Securities and Markets

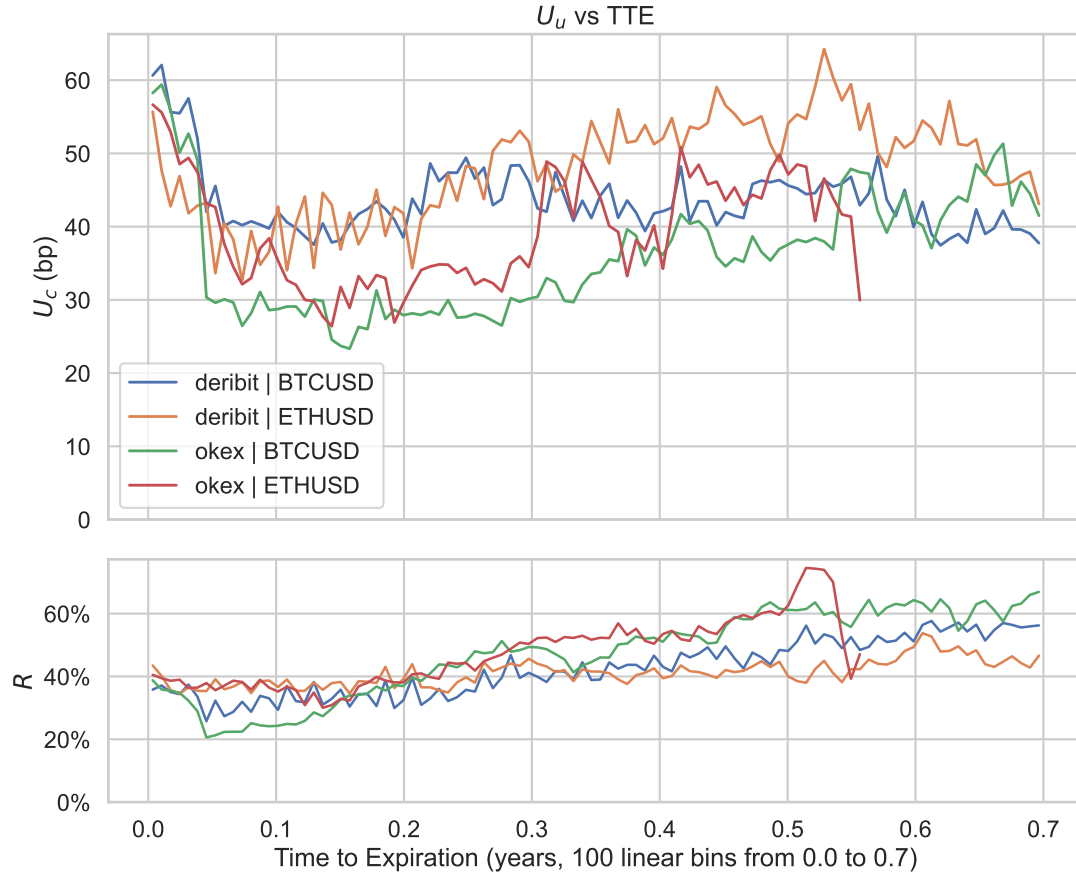


Figure 8: U_u by time to expiry across exchanges and underlyings: the conditional size (upper panel, comparatively flat) and the unfairness rate R (four paths per matched quote) (lower panel). The frequency—and hence the unconditional unfairness—is U-shaped: elevated at the shortest and longest maturities, lowest around one month.

Authority 2024). None of this changed put–call parity; it changed who can hold, finance, and hedge crypto exposure at scale, and under what legal certainty.

Bigger, regulated players plausibly need a smaller premium.

This is where market unfairness is useful as a diagnostic. Whatever keeps a parity wedge open, it is plausibly cheaper to bear for a large, diversified, well-capitalized institution than for the crypto-native firms that dominated before: regulation eases custody and legal risk, balance-sheet depth eases funding risk, diversification eases inventory and convergence risk. If 2024 brought such participants into quoting these markets—directly, or by financing those who do—one would expect the wedge to shrink, beyond any gain in the observable frictions. This is a working hypothesis, not something we identify cleanly: we treat 2024 as a descriptive regime marker and ask whether the evidence fits.

This gives two descriptive questions.

Two descriptive questions follow: did U_u drop around the 2024 changes, and does the drop line up with the observable frictions (spreads, depth, staleness), or persist beyond them? Sections 7 and 8 take these in turn.

7 Did the absolute market unfairness fall in 2024?

7.1 Time-series and distributional evidence

Daily U_u is high before 2024, low after.

The time-series evidence is direct. Figure 9 plots daily U_u by market against the BTCUSD index. Through 2020–2023 U_u is elevated and volatile, spiking in turbulent episodes; from 2024 it drops to a lower, tighter regime. The spikes cluster in stressed markets, when market making is balance-sheet intensive and quote quality poor.

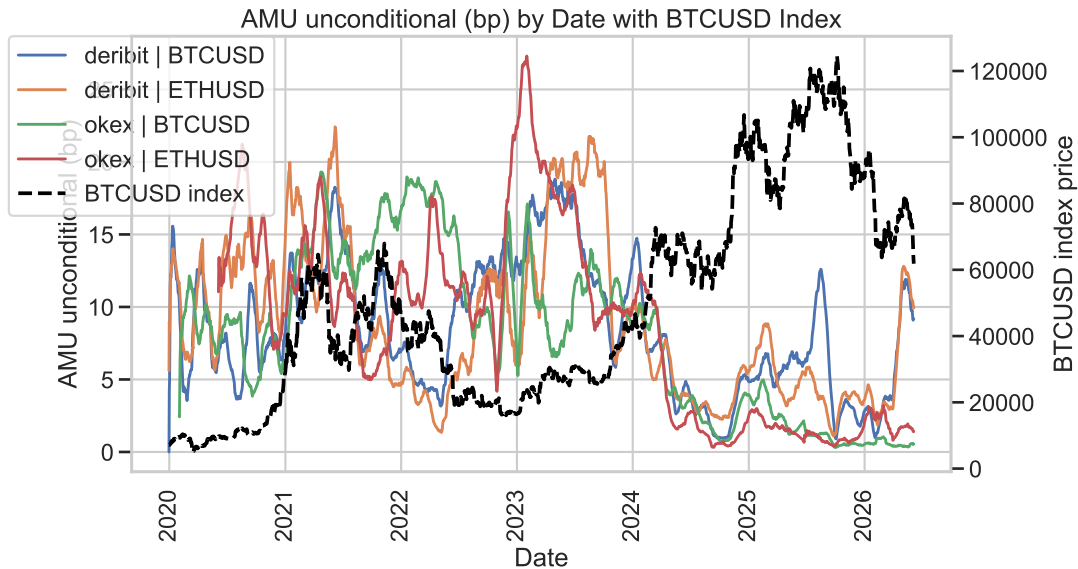


Figure 9: Daily unconditional unfairness by exchange and underlying, 2020–2026, with the BTCUSD index overlaid. It is high and volatile before 2024 and compresses to a lower, tighter band afterward.

The step down tracks the 2024 event dates.

Zooming in, Figure 10 tracks the all-market average across 2023–2024 with event dates

marked. U_u runs high through 2023, steps down around the January 2024 ETP approval, and keeps falling through the Ether ETF and MiCA milestones. The event-study windows (Figure 11) are noisier—the windows overlap—but show the same tendency: lower and less spiky after events.

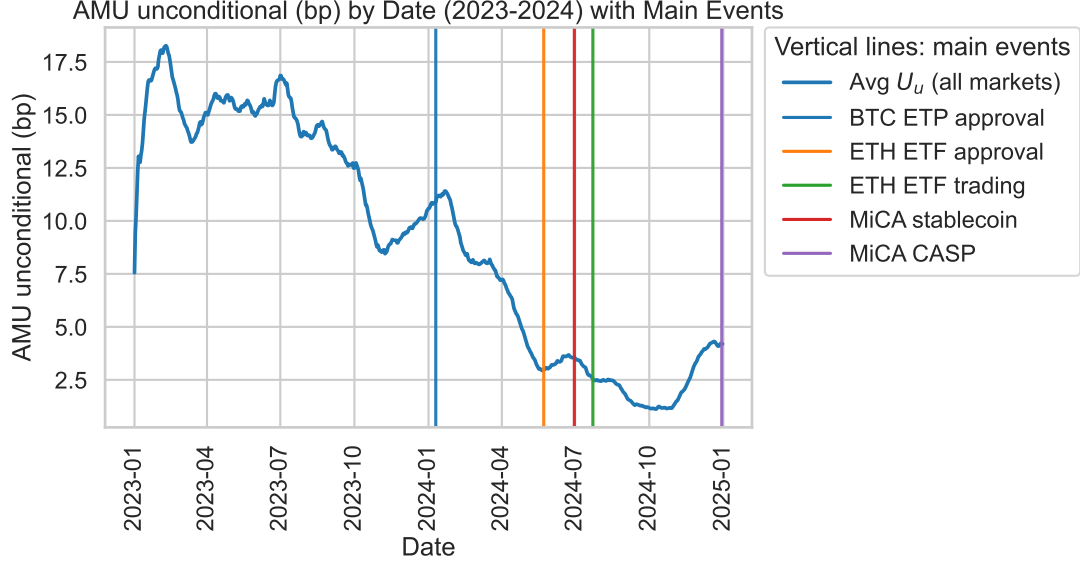


Figure 10: All-market average unconditional unfairness, 2023–2024, with the main institutional event dates marked by vertical lines.

The whole distribution concentrated at the cost floor.

Two figures show *how* the compression happened. Figure 12 overlays the pre- and post-2024 MMA histograms: the post-2024 density concentrates at the cost floor with thinner tails, and average unconditional unfairness falls from 11.78 bp to 4.43 bp. This is consistent with the mechanism of Section 5.1: denser competition erases more of the transient tails. Figure 14 shows the compression over the strike–maturity grid: the pre-2024 hot band at long maturities and away-from-money strikes cools sharply, leaving only the deep-OTM, long-dated corner elevated. Figure 15 shows the change is broad but uneven: conditional unfairness fell in every segment, most in the far-expiry bucket, least on Deribit, more for ETH than BTC and for non-ATM than ATM—largest where the pre-2024 wedge was largest.

The drop does not depend on the cost assumption.

The compression is not an artifact of the cost assumption. Figure 13 evaluates both unfairness flavours over a grid of costs, pre and post. The pre-to-post drop in unconditional unfairness declines only gradually with the assumed cost, staying large across the grid; the drop in conditional unfairness falls away much faster.

8 Regression analysis

8.1 Empirical design

The outcome is cell-level unconditional unfairness.

The dependent variable in all four regressions is the cell-level unconditional unfairness, $U_{u,g,t} = R_{g,t} U_{c,g,t}$ (Eq. 5)—a quote-level, panel-observable object. Regressors enter in three

Event-study windows around 2024 regime markers

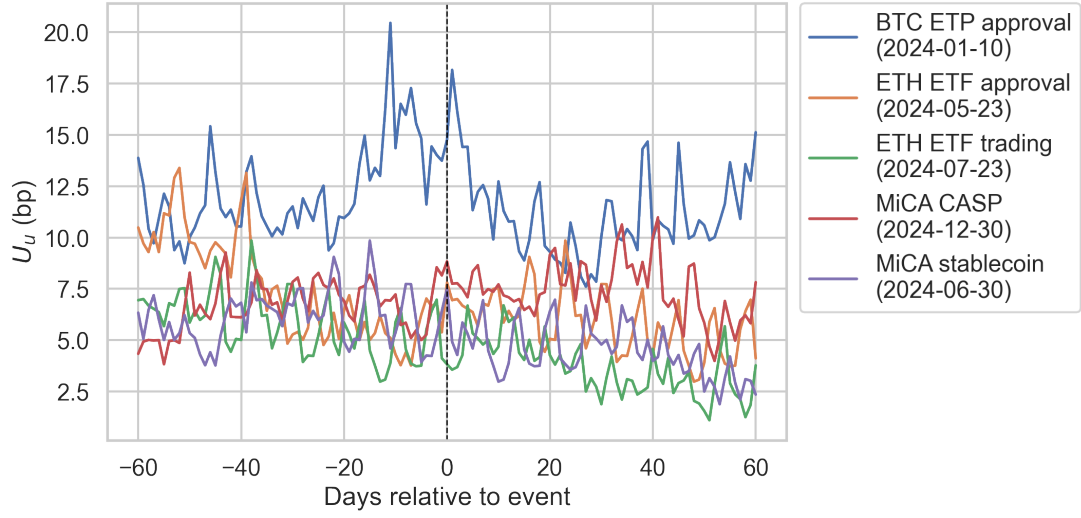


Figure 11: Event-study windows for the unconditional unfairness around the main 2024 institutional event dates, shown separately by event.

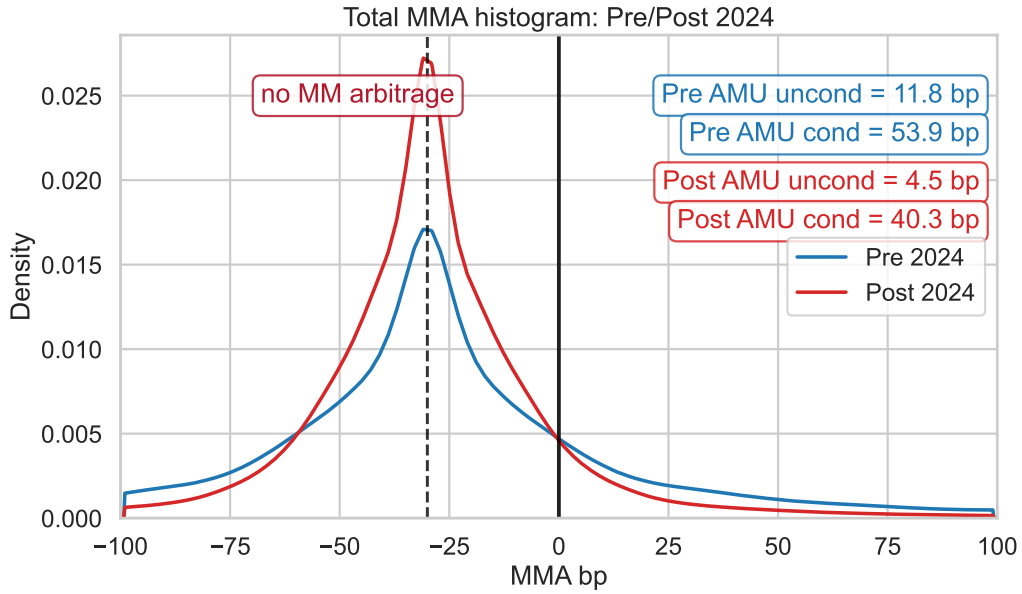


Figure 12: Total MMA histogram across all four paths, split into pre- and post-2024 periods. The post-2024 distribution is more concentrated at the cost floor, and the positive-tail averages—both the conditional and the unconditional unfairness, annotated on the figure—fall markedly after 2024.

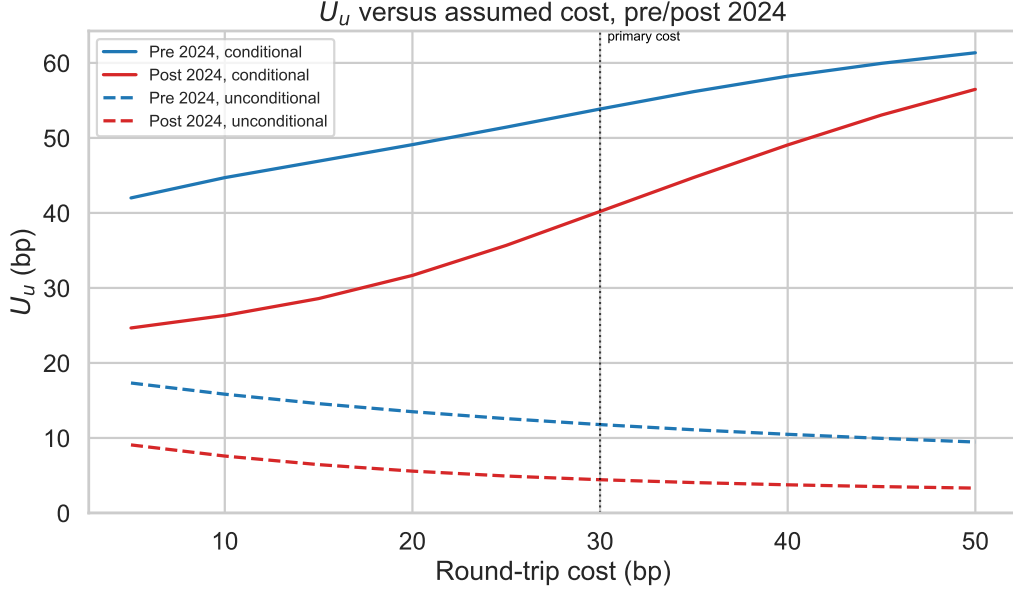


Figure 13: Conditional (solid) and unconditional (dashed) unfairness versus the assumed round-trip cost, pre- (blue) versus post-2024 (red), computed on the same filtered panel as the regressions, so the primary-cost point equals the headline pre/post U_u ; the dotted line marks the primary cost (30 bp). The pre-to-post gap in unconditional unfairness declines only gradually with cost and stays large across the grid; the gap in conditional unfairness shrinks sharply as cost rises. The unconditional measure is therefore the natural object for the regime comparison.

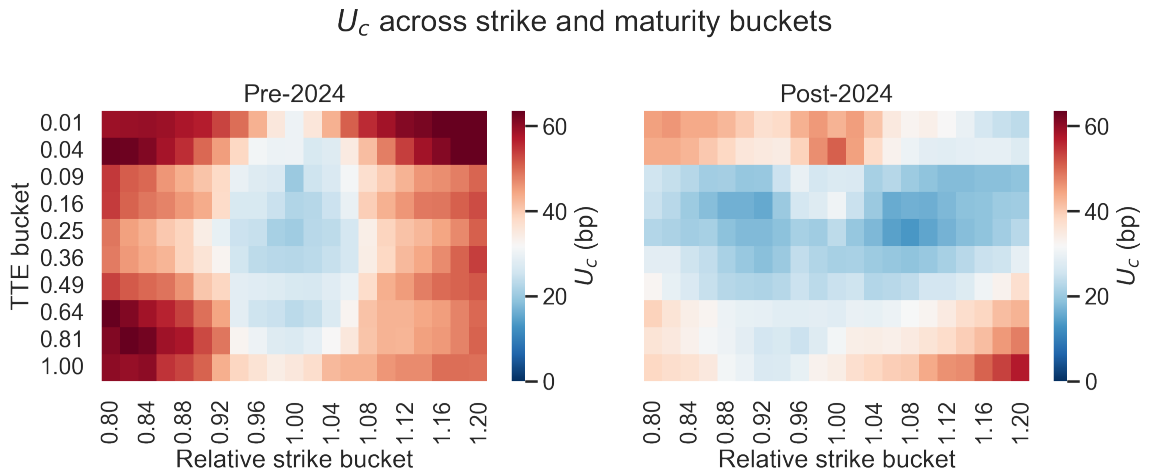


Figure 14: Mean *conditional* U_u across relative-strike and maturity buckets, pre- versus post-2024. The high- U_u interior of the surface cools after 2024; only the deep-OTM, long-dated corner stays elevated.

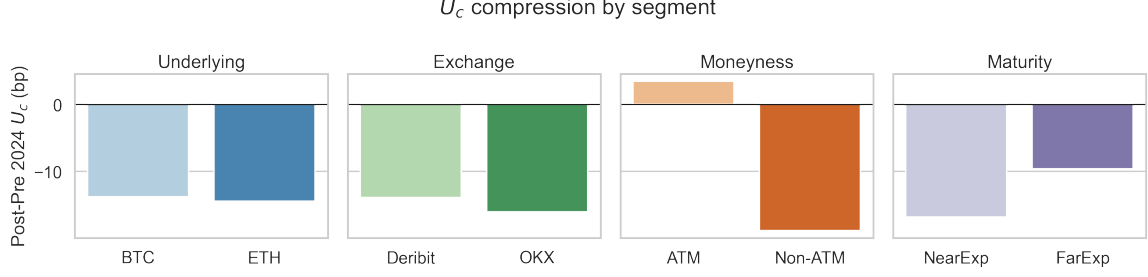


Figure 15: Pre- to post-2024 change in *conditional* unfairness U_c by segment (BTC/ETH, Deribit/OKX, ATM/Non-ATM, NearExp/FarExp). Every segment compresses; the far-expiry bucket compresses most.

blocks: the post-2024 indicator Post_t ; segment dummies (exchange, underlying, moneyness, maturity); and three liquidity frictions,

$$\text{Spr}_{g,t} = \frac{1}{2} \left(\text{mean_call_spread_bp}_{g,t} + \text{mean_put_spread_bp}_{g,t} \right), \quad (7)$$

$$\text{Depth}_{g,t} = \log \left(\text{mean_min_quote_size_dollar}_{g,t} \right), \quad (8)$$

$$\text{Stale}_{g,t} = \text{mean stale-quote fraction across call, put, and spot legs}. \quad (9)$$

Frictions enter as descriptors, not to identify a premium.

The direct-cost term is already subtracted inside every path payoff (Table 1), so market unfairness is the wedge that survives above that floor (Eq. 1). We add the friction variables not to identify a structural premium but to describe: how much of U_u , and of its 2024 change, moves with observable book quality—wider spreads, thinner depth, staler quotes (Amihud & Mendelson 1986, Shleifer & Vishny 1997, Gârleanu & Pedersen 2011, Bollen et al. 2004)—and how much does not.

The break date is January 10, 2024.

Post_t equals one on and after January 10, 2024—the Bitcoin ETP approval, first of the 2024 milestones—treated as a single regime date rather than separate shocks.

We run four nested specifications.

We estimate four specifications, adding regressors in blocks:

$$U_{u,g,t} = \beta_{\text{Post}} \text{Post}_t + \varepsilon_{g,t}, \quad (10)$$

$$U_{u,g,t} = \beta_{\text{Post}} \text{Post}_t + \gamma_g + \varepsilon_{g,t}, \quad (11)$$

$$U_{u,g,t} = \beta_{\text{Post}} \text{Post}_t + \beta_{\text{OKX}} \text{OKX}_g + \beta_{\text{ETH}} \text{ETH}_g + \beta_{\text{ATM}} \text{ATM}_g + \beta_{\text{NE}} \text{NearExp}_g + \varepsilon_{g,t}, \quad (12)$$

$$U_{u,g,t} = \beta_{\text{Post}} \text{Post}_t + \beta_{\text{Depth}} \text{Depth}_{g,t} + \beta_{\text{Spr}} \text{Spr}_{g,t} + \beta_{\text{Stale}} \text{Stale}_{g,t} + \gamma_g + \varepsilon_{g,t}, \quad (13)$$

all with a constant and two-way cluster-robust (by cell and day) standard errors. Specification (1) is the raw regime effect. Specification (2) adds cell fixed effects γ_g , isolating the within-cell shift. Specification (3) instead adds the segment dummies as pooled OLS: there the dummies are the objects of interest (the cross-sectional question), so we estimate rather than absorb them. Specification (4) adds the frictions with cell fixed effects. We prefer cell fixed effects because the surface dependence is strongly nonlinear (Figures 7–8); linear dummies would leave segment structure in the residual. In (4), β_{Post} is the within-cell change net of both the frictions and the full segment structure. Table 5 defines the abbreviations.

Table 5: Abbreviations used in the empirical specifications.

Abbreviation	Definition
Post_t	Post-2024 regime indicator (1 for dates on/after 2024-01-10, 0 otherwise).
<i>Liquidity-friction variables</i>	
$\text{Spr}_{g,t}$	Average put-call spread in basis points, $0.5 \times (\text{mean_call_spread_bp} + \text{mean_put_spread_bp})$.
$\text{Depth}_{g,t}$	Log quote-depth proxy, $\log(\text{mean_min_quote_size_dollar})$, min taken over put/call bid/ask quote sizes.
$\text{Stale}_{g,t}$	Mean stale-quote fraction across call, put, and spot legs. A stale flag does not by itself mean a quote is wrong or unfillable: it can equally mean the market has not moved enough to warrant a requote, so a correctly-priced quote may sit unchanged for long stretches. It is a proxy for update intensity, not for quote validity.
<i>Segment dummies (components of the cell fixed effect γ_g)</i>	
OKX_g	OKX indicator (1 for OKX, 0 for Deribit).
ETH_g	ETH-underlying segment indicator.
ATM_g	At-the-money indicator, $ \text{rel_strike_bucket} - 1 \leq 0.05$.
NearExp_g	Near-expiry indicator, $\text{tte_bucket} \leq 0.35$ years.
γ_g	Cell fixed effect for the tuple $(\text{exchange}, \text{ref_sym}, \text{rel_strike_bucket}, \text{tte_bucket})$.
$\varepsilon_{g,t}$	Idiosyncratic error.

8.2 Results

We use the regressions to check that market unfairness behaves like a market-quality statistic.

We do not try to identify why U_u fell. We ask whether it behaves as a good market-quality measure should. Two things are required: it should fall when the market is independently believed to have improved (the 2024 institutionalization), and that fall should not be an artifact of which contracts trade. Table 6 checks both. The columns are (1) the raw post-2024 shift; (2) plus cell fixed effects; (3) segment dummies (pooled OLS); (4) all three frictions with cell fixed effects; and (5)–(7) each friction on its own, with cell fixed effects. Each cell stacks three rows: the coefficient (with significance stars), the two-way cluster-robust (by cell and day) standard error in parentheses, and, in bold brackets, the standardized effect $\beta \text{std}(X)/\text{std}(Y)$.

U_u is markedly lower after 2024.

Specification (1). The post-2024 coefficient is negative and large, and stays highly significant under two-way clustered errors. U_u is much lower after 2024—the period the market is widely agreed to have improved. This is the first thing a market-quality measure should show, and addresses the second question: U_u did drop upon the 2024 regime shift.

The drop is within cells, not a change in contract mix.

Specification (2). Adding cell fixed effects barely moves the post-2024 coefficient. The drop happens *within* fixed strike–maturity–venue cells over time, not because the traded mix shifted. This rules out the main composition artifact—the second thing we need.

U_u differs across segments, but only slightly.

Specification (3). With segment dummies (pooled OLS), U_u differs by exchange and moneyiness—both strongly significant even under clustered errors—while the underlying (ETH) and near-expiry effects are marginal or insignificant once errors are clustered. All are small next to variation over the surface. The low R^2 is expected: linear dummies cannot capture the nonlinear surface, which is why (4) uses fixed effects. This addresses the first question: U_u differs across venues and underlyings, but only modestly.

The three frictions together explain the drop.

Specification (4). We add spread, depth, and staleness, still with cell fixed effects. All three are strong under two-way clustered errors (by cell and by day). Together they pull the post-2024 coefficient down to about zero, and it is no longer significant. So on the filtered panel these three book-quality measures fully account for the within-cell drop.

No single friction does it on its own.

Specifications (5)–(7). Here we add the frictions one at a time. With depth alone (5), or staleness alone (7), the post-2024 coefficient barely moves and stays large and significant. With spread alone (6) it falls a lot but is still significant. Only all three together, in (4), bring it to zero. So the drop is a mix of the three frictions, and spread does most of the work.

Is the compression fewer wedges, or smaller ones?

The regime effect has two margins. Because the per-quote wedge factors as $U_u = R U_c$, regressing $\log R$ and $\log U_c$ on the post-2024 indicator with cell fixed effects splits the log change in unconditional unfairness *exactly* into an extensive margin (how often an executable wedge exists) and an intensive margin (how large it is when it does): the two semi-elasticities sum to the unconditional one. Table 7 reports the split over the within-cell, positive-wedge sample.

Term	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post _t	-8.444*** (0.395) [-29%]	-8.953*** (0.346) [-31%]	-8.813*** (0.369) [-30%]	0.846*** (0.149) [3%]	-7.574*** (0.313) [-26%]	-0.153 (0.171) [-1%]	-6.966*** (0.326) [-24%]	
OKX _g			-1.346*** (0.415) [-5%]					
ETH _g			0.675* (0.389) [2%]					
ATM _g			-9.800*** (0.374) [-30%]					
NearExp _g			-0.720* (0.381) [-2%]					
Depth _{g,t}				-0.848*** (0.055) [-12%]	-1.490*** (0.112) [-20%]			-0.710*** (0.060) [-10%]
Spr _{g,t}				0.158*** (0.001) [80%]		0.163*** (0.001) [83%]		0.156*** (0.001) [79%]
Stale _{g,t}				9.879*** (0.788) [7%]			40.496*** (1.081) [28%]	9.483*** (0.791) [7%]
Total R ²	0.085	0.294	0.178	0.763	0.304	0.756	0.359	0.763
Observations	738,310	738,310	738,310	738,310	738,310	738,310	738,310	738,310
Fixed effects	None	cell_id	None	cell_id	cell_id	cell_id	cell_id	cell_id

Table 6: Regressions of the cell-level unconditional unfairness. Column (1) regresses U_u on the post-2024 indicator; (2) adds cell fixed effects to (1), isolating the within-cell regime effect; (3) instead adds the exchange, underlying, moneyness, and maturity dummies (pooled OLS); (4) adds the depth, spread, and staleness controls with cell fixed effects; (5)–(7) add the frictions one at a time (depth, spread, staleness), each with cell fixed effects. Each coefficient cell stacks three rows: the coefficient, the two-way cluster-robust (by cell and day) standard error in parentheses, and, in bold brackets, the standardized effect $\beta \text{std}(X)/\text{std}(Y)$ (the change in U_u , in standard deviations of U_u , per one-standard-deviation increase in the regressor). Stars denote p -value bands from the t -statistic. Standard errors are two-way clustered, by cell and by day.

	Unfairness rate R	Cond. size U_c	U_u
Channel			
Post _{t}	-1.160***	-0.602***	-1.762***
Total R^2	0.395	0.150	0.334
Observations	613,730	613,730	613,730

Table 7: Two-part (extensive/intensive) decomposition of the post-2024 U_u compression. Each column regresses the log of one channel—the unfairness rate R , the conditional wedge size U_c , and their product the unconditional unfairness U_u —on the post-2024 indicator with cell fixed effects, over the common sample of cell-days that carry a positive wedge. Entries are the Post coefficients (in log points, i.e. semi-elasticities: approximately the fractional change in the channel), with significance stars from two-way cluster-robust (by cell and day) t -statistics. Because $\log U_u = \log R + \log U_c$ row by row, the frequency and size coefficients sum to the unconditional coefficient by construction.

U_u tracks the spread channel tightly.

Figure 16 shows how closely the wedge moves with spread—which is why spread does most of the work in specification (4).

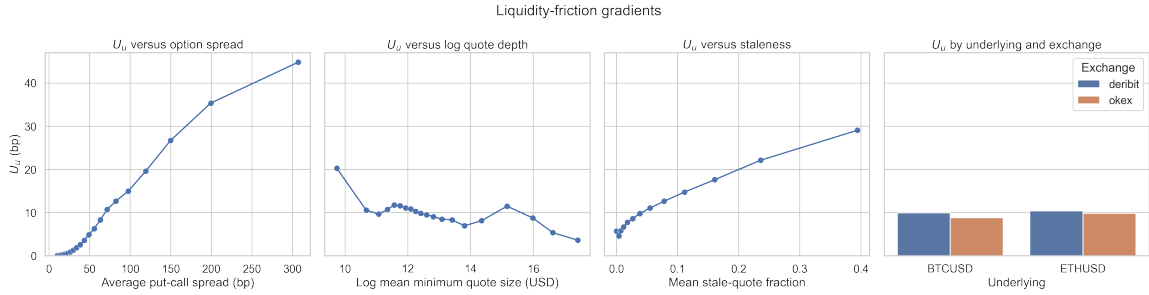


Figure 16: Unconditional unfairness against average put–call spread, log quote depth, and stale-quote fraction, and by underlying and exchange. U_u rises steeply with the spread and with staleness—the friction channels that specification (4) controls for—while its dependence on depth is weak and non-monotone. Cross-sectionally it is slightly higher on OKX than Deribit and for ETH than BTC, echoing the segment dummies in specification (3).

U_u behaves like a market-quality measure.

Put together: U_u differs only a little across segments; it is much lower after 2024; and that drop is within cells and is explained by the frictions, mostly spread. So U_u moves with book quality, and it fell when the market is agreed to have improved. We do not claim market unfairness adds information beyond the frictions, and we do not identify which participants drove the change.

9 Robustness

Standard errors are clustered, not treated as independent.

The regression uses one observation per cell-day, but these are not independent draws. A cell—a fixed strike/maturity/venue bucket—is observed on many days and its wedge is per-

sistent, so its residuals are correlated over time; and all cells share market-wide conditions on a given day, so residuals are correlated across cells within a date. Heteroskedasticity-robust (HC1) errors assume independence across cell-days and are, in consequence, far too small: with hundreds of cells observed over hundreds of days, the effective amount of independent information is closer to the number of cells or days than to the number of cell-days. We therefore report two-way cluster-robust (by cell and day) standard errors, clustering two ways—by cell and by day (Cameron et al. 2011). Only the covariance changes; the coefficients are identical. The mechanism is that the clustered variance sums each cluster’s aggregated score *before* squaring it, so positively correlated within-cluster residuals no longer cancel. In our panel the clustered errors are several times larger than HC1; that is what the reported standard errors and significance stars reflect. The 2024 compression and the friction gradients remain significant under this more conservative variance, but their t -statistics are far smaller than a naive HC1 calculation would suggest.

The friction controls enter jointly, and collinearity does not distort the regime effect.

Spread, depth, and staleness are strongly correlated—a thin book tends to be wide, shallow, and slow to update at once. Collinearity of this kind inflates the variances of the *individual* friction coefficients and makes them hard to read one by one, but it does not bias the coefficient of interest, the post-2024 indicator, which is identified off variation orthogonal to the frictions. We verified this with a principal-component check: because the principal components are an orthogonal rotation of the three standardized frictions, entering all three components reproduces specification (4) exactly, and any full-rank recombination of the frictions leaves the post-2024 coefficient unchanged. (A single component is not a substitute: keeping only the first principal component is a rank-one control that discards the other two friction directions—one of which is strongly related to U_u despite carrying little of the frictions’ variance—so it merely under-controls and inflates the apparent regime effect.) We therefore report the three frictions jointly in specification (4).

The results do not hinge on the cost or the discount rate.

Two auxiliary parameters enter the construction: the round-trip cost (baseline 30 bp) and the discount rate (baseline $r = 5\%$). Neither is crucial. The cost only sets the scale of market unfairness—a fixed proportional cost shifts the level, but the pre/post *difference* in unconditional unfairness barely moves with it (the conditional form is more cost-sensitive, one reason the headline uses the unconditional measure), and a common cost keeps cross-venue comparisons coherent. The discount rate matters even less, for a structural reason: r enters only through e^{-rT} , which multiplies a small quantity—the futures leg $(F - K)$ in each payoff and the futures half-spread $\frac{1}{2}(F_{ask} - F_{bid})$ in cost’. Its effect on a payoff is $\partial[(F - K)e^{-rT}] / \partial r = -T(F - K)e^{-rT}$, of order $T(F - K)/\text{index}$; since our strikes sit within $\pm 21\%$ of the forward and most observation mass sits in near-the-money, short-dated cells where $(F - K)$ and T are smallest, plausible changes in r move U_u negligibly—and the pre/post difference more negligibly still, since r shifts both regimes almost identically. Evaluating U_u across grids of both parameters—the cost grid read off the same filtered panel as the regressions, the discount-rate grid recomputed at each r (Table 8)—bears this out. A higher assumed cost lowers the U_u level and shrinks the compression somewhat, but the pre/post drop stays large and negative across the whole cost grid; across r it is essentially constant. Under every choice the 2024 compression is present and sizable.

The compression survives every reasonable filter choice.

A final check varies the construction one choice at a time—the positive-part clip, the

cost (bp)	pre	post	Δ		r	pre	post	Δ
10	15.83	7.59	-8.25		0.000	12.90	5.66	-7.24
20	13.50	5.58	-7.92		0.025	12.57	4.92	-7.65
30	11.78	4.43	-7.35		0.050	12.39	4.46	-7.93
40	10.49	3.75	-6.73		0.075	12.39	4.32	-8.07
50	9.46	3.31	-6.14		0.100	12.55	4.53	-8.02

Table 8: U_u pre- and post-2024, in basis points of notional, across a grid of round-trip costs (left; read off the same filtered panel as the regressions, so the baseline-cost column matches the headline pre/post U_u) and of discount rates r (right; recomputed at each rate; baseline 5%). Both grids share one estimand at their baselines. Δ is the post–pre change.

stale-quote filter, the spread and size cuts, the moneyiness window, and tick- versus notional-weighting—recomputing the pre/post unconditional unfairness for each (Table 9). The level moves with the filters, as it must, but the post-2024 drop is large and negative under every setting; none of the choices manufactures the compression or erases it.

setting	pre	post	Δ
baseline	11.78	4.50	-7.28
clip 50 bp	7.47	3.02	-4.45
clip 200 bp	16.65	6.44	-10.22
no clip	19.11	8.20	-10.91
drop stale quotes	9.04	3.85	-5.19
spread cap 500 bp	9.87	3.34	-6.53
spread cap 200 bp	4.19	2.06	-2.13
min size \$25k	11.28	4.15	-7.12
moneyiness $\pm 10\%$	6.66	3.13	-3.54
all strikes	18.31	10.15	-8.16
notional-weighted	7.97	2.28	-5.69

Table 9: Robustness of the pre/post-2024 unconditional unfairness (bp of notional) to the filter and aggregation choices, each varied one at a time from the baseline (top row): the positive-part clip, the stale-quote filter, the spread and minimum-size cuts, the moneyiness window, and notional- versus tick-weighting. U_u is pooled over ticks and the four paths at the baseline cost, over the same tick-level filtered sample the panel is built from, so the baseline row (top) reproduces the headline pre/post U_u ; the invariant of interest is the stability of the pre/post drop across settings. Δ is the post–pre change.

10 Conclusion

U_u is a model-free symptom of imperfect quoting.

We propose *Absolute Market Unfairness*: the cost-adjusted put–call-parity violation left in the standing quotes of inverse cryptocurrency option markets. It is model-free and comparable in basis points—a transparent sign that the top of the book is not where

frictionless parity would put it—not a full account of a venue’s liquidity or a cleanly identified risk premium. It says nothing about fundamental mispricing and guarantees no dollar profit.

U_u is small across venues, shaped by the surface, and fell in 2024.

Using Deribit and OKX BTCUSD and ETHUSD options from 2020 through June 2026: U_u differs across venues and underlyings, but little next to its variation over the surface, where it is governed mainly by how often a wedge appears—smallest for month-scale, near-the-money contracts, largest in the illiquid corners. It fell sharply over 2023–2024, around the ETF and MiCA dates—though much of the decline was already under way before them—and the compression is present at every cost level we examine; most of the drop survives controls for spreads, depth, and staleness within cells. This is consistent with the entry of participants whose quoting improved market quality, not just more passive depth—though the design is descriptive and does not identify the mechanism. The value of market unfairness is that it turns a qualitative worry about quote quality into a model-free number, comparable across venues, underlyings, and maturities, and trackable through regulatory change.

The three questions, answered.

How large is U_u , and does it differ across exchanges and underlyings? It is small—about 7.22 bp of notional on average (11.78 bp before 2024, 4.43 bp after)—and it differs across venues and underlyings only modestly; far more of its variation comes from position on the option surface. *Did U_u drop upon the 2024 regulatory and institutional regime shift?* Yes: it fell sharply over 2023–2024 around the ETF and MiCA dates (much of the decline preceding them), the drop is highly significant under standard errors clustered by cell and day, and it is present at every trading cost and discount rate we examine. *Is the drop fully explained by the usual liquidity frictions, or does it partly reflect a new class of parity-enforcing participants?* It is not fully explained by spreads, depth, and staleness: a substantial within-cell drop survives them. This is consistent with—though our descriptive design does not identify—the entry of participants whose quoting improved market quality rather than merely adding passive depth.

References

- Alexander, C., Chen, D. & Imeraj, A. (2023), ‘Crypto quanto and inverse options’, *Mathematical Finance* **33**(4), 1005–1043.
- Alexander, C., Chen, X., Deng, J. & Wang, T. (2024), ‘Arbitrage opportunities and efficiency tests in crypto derivatives’, *Journal of Financial Markets* **71**, 100930.
- Amihud, Y. (2002), ‘Illiquidity and stock returns: cross-section and time-series effects’, *Journal of Financial Markets* **5**(1), 31–56.
- Amihud, Y. & Mendelson, H. (1986), ‘Asset pricing and the bid-ask spread’, *Journal of Financial Economics* **17**(2), 223–249.
- Bollen, N. P. B., Smith, T. & Whaley, R. E. (2004), ‘Modeling the bid/ask spread: measuring the inventory-holding premium’, *Journal of Financial Economics* **72**(1), 97–141.
- Cameron, A. C., Gelbach, J. B. & Miller, D. L. (2011), ‘Robust inference with multiway clustering’, *Journal of Business & Economic Statistics* **29**(2), 238–249.

- Chen, S., Chng, M. T. & Liu, Q. (2021), ‘The implied arbitrage mechanism in financial markets’, *Journal of Econometrics* **222**(1), 468–483.
- Cremers, M. & Weinbaum, D. (2010), ‘Deviations from put-call parity and stock return predictability’, *Journal of Financial and Quantitative Analysis* **45**(2), 335–367.
- Degryse, H., de Jong, F., van Ravenswaaij, M. & Wuyts, G. (2005), ‘Aggressive orders and the resiliency of a limit order market’, *Review of Finance* **9**(2), 201–242.
- Demsetz, H. (1968), ‘The cost of transacting’, *The Quarterly Journal of Economics* **82**(1), 33–53.
- Du, W., Tepper, A. & Verdelhan, A. (2018), ‘Deviations from covered interest rate parity’, *The Journal of Finance* **73**(3), 915–957.
- El-Mekkaoui, M. & Flood, M. D. (1998), ‘Put-call parity revisited: Intraday evidence from the options market’, *Journal of International Financial Markets, Institutions and Money* **8**(3–4), 357–376.
- European Securities and Markets Authority (2024), ‘Markets in crypto-assets regulation (mica)’, Regulatory information page. MiCA became applicable in stages in 2024. Accessed 2026-06-04.
URL: <https://www.esma.europa.eu/esmas-activities/digital-finance-and-innovation/markets-crypto-assets-regulation-mica>
- Felföldi-Szűcs, N., Králik, B. & Váradi, K. (2024), ‘Put–call parity in a crypto option market — evidence from binance’, *Finance Research Letters* **61**, 104874.
- Foucault, T., Pagano, M. & Röell, A. (2013), *Market liquidity: theory, evidence, and policy*, Oxford University Press, USA.
- Gagnon, L. & Karolyi, G. A. (2010), ‘Multi-market trading and arbitrage’, *Journal of Financial Economics* **97**(1), 53–80.
- Gârleanu, N. & Pedersen, L. H. (2011), ‘Margin-based asset pricing and deviations from the law of one price’, *The Review of Financial Studies* **24**(6), 1980–2022.
- Glosten, L. R. & Milgrom, P. R. (1985), ‘Bid, ask and transaction prices in a specialist market with heterogeneously informed traders’, *Journal of financial economics* **14**(1), 71–100.
- Gromb, D. & Vayanos, D. (2010), ‘Limits of arbitrage’, *Annual Review of Financial Economics* **2**(1), 251–275.
- Ho, T. & Stoll, H. R. (1981), ‘Optimal dealer pricing under transactions and return uncertainty’, *Journal of Financial Economics* **9**(1), 47–73.
- Kamara, A. & Miller, T. W. (1995), ‘Daily and intradaily tests of european put-call parity’, *The Journal of Financial and Quantitative Analysis* **30**(4), 519–539.
- Klemkosky, R. C. & Resnick, B. G. (1979), ‘Put-call parity and market efficiency’, *The Journal of Finance* **34**(5), 1141–1155.

- Králik, B. (2026a), ‘publications: Amu-liquidity-premium’, GitHub repository. Accessed 2026-07-16.
URL: <https://github.com/balazs14/publications/tree/main/AMU-liquidity-premium>
- Králik, B. (2026b), ‘Tardisproject’, GitHub repository. Accessed 2026-07-16.
URL: <https://github.com/balazs14/TardisProject>
- Krishnamurthy, A. (2002), ‘The bond/old-bond spread’, *Journal of Financial Economics* **66**(2–3), 463–506.
- Kyle, A. S. (1985), ‘Continuous auctions and insider trading’, *Econometrica* **53**(6), 1315–1335.
- Large, J. (2007), ‘Measuring the resiliency of an electronic limit order book’, *Journal of Financial Markets* **10**(1), 1–25.
- Lucic, V. & Sepp, A. (2024), ‘Valuation and hedging of cryptocurrency inverse options’, *Quantitative Finance* .
- Nisbet, M. (1992), ‘Put-call parity theory and an empirical test of the efficiency of the london traded options market’, *Journal of Banking & Finance* **16**(2), 381–403.
- Ofek, E., Richardson, M. & Whitelaw, R. F. (2004), ‘Limited arbitrage and short sales restrictions: Evidence from the options markets’, *Journal of Financial Economics* **74**(2), 305–342.
- Pan, K. & Zeng, Y. (2017), Etf arbitrage under liquidity mismatch, Working Paper 59, European Systemic Risk Board.
- Pontiff, J. (1996), ‘Costly arbitrage: Evidence from closed-end funds’, *The Quarterly Journal of Economics* **111**(4), 1135–1151.
- Pontiff, J. (2006), ‘Costly arbitrage and the myth of idiosyncratic risk’, *Journal of Accounting and Economics* **42**(1–2), 35–52.
- Shleifer, A. & Vishny, R. W. (1997), ‘The limits of arbitrage’, *The Journal of Finance* **52**(1), 35–55.
- Tardis.dev (2026), ‘Tardis: Historical crypto market data api’, Product and documentation website. Accessed 2026-06-16.
URL: <https://tardis.dev/>
- Todorov, K. (2021), ‘The anatomy of bond etf arbitrage’, *BIS Quarterly Review* pp. 41–53.
- U.S. Securities and Exchange Commission (2005), ‘Regulation nms, release no. 34-51808: Order protection (trade-through) rule’, Final Rule, 17 CFR Parts 200, 201, 230, 240, 242, 249, and 270. Accessed 2026-07-16.
URL: <https://www.sec.gov/rules/final/34-51808.pdf>
- U.S. Securities and Exchange Commission (2024a), ‘Order granting accelerated approval of proposed rule changes to list and trade shares of ether-based exchange-traded products’, SEC Release, May 2024. Accessed 2026-06-04.
URL: <https://www.sec.gov/>

U.S. Securities and Exchange Commission (2024*b*), ‘Statement on the approval of spot bitcoin exchange-traded products’, SEC Chair Statement, January 10, 2024. Accessed 2026-06-04.

URL: *<https://www.sec.gov/newsroom/speeches-statements/gensler-statement-spot-bitcoin-011023>*